

Research Paper

Public procurement risk indicators from open government data and provenance model

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Abstract. Public procurement (PP) auditing increasingly relies on Open Government Data (OGD) to detect fraudulent patterns and support oversight activities. However, transforming dispersed and heterogeneous government data into structured, auditable, and practically useful risk indicators remains a significant challenge. This study proposes a method for constructing open risk indicators for PP based on the integration of government datasets and provenance modeling. The method combines knowledge elicitation from academic literature and domain experts with a structured data and analytics pipeline that operationalizes heterogeneous OGD into measurable indicators while explicitly documenting their derivation processes. The method was instantiated in the Brazilian context, integrating multiple public datasets and resulting in the construction of 26 risk indicators grounded in both literature and auditing practice. A qualitative evaluation was conducted with 28 public auditors, who assessed the relevance of the datasets and indicators for procurement oversight. The results show strong agreement among practitioners regarding the usefulness of the proposed indicators, particularly when supported by integrated datasets. Additionally, the inclusion of provenance modeling was perceived as a valuable mechanism to enhance transparency, traceability, and trust in data-driven auditing processes. These findings demonstrate that combining OGD integration with provenance-aware modeling enables the development of interpretable and auditable risk indicators, contributing to more reliable and transparent decision-support tools for public procurement oversight.

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1. Introduction

Public procurement (PP) refers to the process through which governments acquire goods, services, and infrastructure from private suppliers to support the provision of public services and the functioning of the state (Telgen et al., 2012). PP represents one of the largest areas of public expenditure worldwide, accounting for up to 25% of Gross Domestic Product (GDP) in Organisation for Economic Co-operation and Development (OECD) countries (Fazekas & Blum, 2021). Across Europe, initiatives such as the EU PP Strategy and the European Single Procurement Document (ESPD) aim to harmonize procedures and enhance integrity (Telles, 2017). Yet procurement systems remain vulnerable to inefficiencies and corruption risks (Sharma et al., 2019). On the other hand, it is possible to observe the scale and complexity to properly monitor and evaluate PPs, like in the Brazilian context: within a highly decentralized federal structure, procurement activities span thousands of

entities at federal, state, and municipal levels (Ribeiro et al., 2018). When combined with open data initiatives, this complexity creates opportunities for transparency while simultaneously raising governance challenges related to accountability and public trust (McCue et al., 2015). In this context, Open Government Data (OGD) has emerged as a key instrument for promoting oversight (Soylu et al., 2022), and analytical techniques are increasingly applied to procurement monitoring (Nai et al., 2022). However, before deploying advanced analytics at scale, it is essential to establish clear, auditable red flags grounded in open data and aligned with auditing practice.

In the context of PP auditing, risk indicators are observable patterns derived from transactional and institutional data that may signal irregularities, conflicts of interest, or vulnerabilities (Felizzola et al., 2024). Rather than constituting direct evidence of wrongdoing, these indicators function as heuristic signals that assist auditors in prioritizing investigations and identifying cases that warrant closer scrutiny. Prior research has explored different types of PP risk indicators, including signals related to limited competition, unusual bidding patterns, etc (Fazekas et al., 2016), (Ferwerda et al., 2017), (Tátrai & Németh, 2018). However, despite their widespread use, the operationalization of such indicators often remains methodologically insufficiently transparent and poorly traceable. Many studies (Fuertes et al., 2025), (Modrušan et al., 2021), (Lisciandra et al., 2022) focus just on statistical or machine learning detection methods applied to specific procurement datasets, often with limited transparency regarding data origin, labeling procedures, and the generation of indicators used for model training. As a result, the analytical pipeline linking raw government data to risk or anomalies detection, among other tasks frequently remains insufficiently documented and difficult to audit.

In the context of PP auditing, risk indicators, also referred in the literature as red flags, are observable patterns derived from transactional and institutional data that may signal irregularities, conflicts of interest, or vulnerabilities (Felizzola et al., 2024). Rather than constituting direct evidence of wrongdoing, these indicators function as heuristic signals that assist auditors in prioritizing investigations and identifying cases that warrant closer scrutiny. Prior research has explored different types of PP risk indicators, including signals related to limited competition, unusual bidding patterns, etc (Fazekas et al., 2016), (Ferwerda et al., 2017), (Tátrai & Németh, 2018). However, despite their widespread use, the operationalization of such indicators often remains methodologically insufficiently transparent and poorly traceable. Many studies (Fuertes et al., 2025), (Modrušan et al., 2021), (Lisciandra et al., 2022) focus just on statistical or machine learning detection methods applied to specific procurement datasets, often with limited transparency regarding data origin, labeling procedures, and the generation of indicators used for model training. As a result, the analytical pipeline linking raw government data to risk or anomalies detection, among other tasks frequently remains insufficiently documented and difficult to audit.

Addressing these limitations requires mechanisms capable of documenting how indicators are constructed from underlying datasets and analytical transformations. In this context, data provenance has emerged as an important mechanism for documenting the origin of data, the transformations applied to it, and the analytical processes involved in generating indicators or models (Curcin, 2017; Herschel et al., 2017). By explicitly recording the relationships between datasets, transformations, and analytical outputs, provenance enables stakeholders to reconstruct the full derivation path from raw data to risk indicators. This capability is particularly relevant in PP auditing, where analytical results must be interpretable and auditable by public institutions. In this study, we adopt a provenance model inspired by the W3C PROV (PROVenance) Data Model (Groth & Moreau, 2013), in which the analytical workflow required to construct each risk indicator is modeled in advance, enabling transparent and reproducible indicator–data–source relationships.

This paper addresses these limitations by proposing a method for constructing open risk indicators for PP based on OGD integration and provenance modeling. The motivation for the study emerged from semi-structured interviews conducted with domain experts from Brazilian public oversight institutions, presented in Section 2, which revealed recurring challenges related to the practical use of analytical tools, the operational interpretation of risk indicators, data access and integration barriers, institutional constraints, and the need for transparent and auditable analytical processes. In particular, auditors emphasized that, although risk indicators are increasingly used to support procurement investigations, their construction frequently lacks explicit traceability regarding the datasets, transformations, and analytical procedures involved. These concerns reinforce limitations also identified in the literature, discussed in Section 3, where existing studies often prioritize statistical or machine learning approaches while providing limited transparency regarding how indicators are operationalized from government data. As a result, the analytical processes connecting raw OGD to fraud risk indicators frequently remain insufficiently documented, difficult to reproduce, and challenging to

audit in institutional contexts. In this context, the study is guided by the following overarching research question: *How can open government data be transformed into transparent and traceable risk indicators for public procurement oversight?* To address this question, the study also considers three complementary subquestions aligned with its main research phases: (RQ1) *What practical challenges affect the use and interpretation of risk indicators in public procurement auditing?*; (RQ2) *How have prior studies defined and operationalized risk indicators for public procurement?*; and (RQ3) *How can provenance modeling support the transparent and traceable construction of risk indicators from heterogeneous government datasets?*

To address these questions, the study adopts a Design Science Research (DSR) approach, described in Section 4, combining empirical insights from auditors with a structured literature review to derive the design requirements of the proposed method. The resulting method, presented in Section 5, integrates heterogeneous government datasets through a structured data and analytics pipeline capable of generating measurable risk indicators while explicitly documenting their derivation processes through provenance modeling. The method was instantiated in the Brazilian PP context using multiple OGD sources, resulting in the construction of 26 risk indicators grounded in both academic literature and auditing practice. Subsequently, a qualitative evaluation, presented in Section 6, was conducted with 28 public auditors to assess the perceived relevance of the datasets and the practical applicability of the proposed indicators in procurement oversight activities. The results demonstrate strong practitioner alignment with the proposed indicators and highlight the relevance of provenance-aware modeling as a mechanism to improve transparency, traceability, and trust in data-driven procurement oversight. Finally, Sections 7 and 8 discuss the implications, contributions, limitations, and future research directions of the study.

2. Empirical Background

This section provides an empirical background that grounds the research problem in real-world auditing practices and informs the subsequent literature review. While prior studies offer conceptual and analytical perspectives on fraud detection in PP, they overlook how risk indicators are interpreted, validated, and operationalized in institutional contexts. The semi-structured interviews conducted with public auditors were therefore used to identify practical challenges—such as data fragmentation, limited interoperability, and the need for explainable analytical outputs—as well as to refine the analytical focus of the study. In particular, insights from practitioners guided the definition of the review scope, supporting the identification of relevant types of risk indicators (or red flags) and emphasizing the importance of transparency and traceability in their construction. As a result, the empirical findings serve as a foundation for the literature review and contribute to the derivation of design requirements for the proposed method.

A total of three interviews were carried out with professionals directly involved in fraud investigation and procurement oversight activities. Two of the interviewees were affiliated with the Administrative Council for Economic Defense (CADE), the Brazilian authority responsible for competition enforcement and the investigation of anticompetitive conduct, including collusion in public procurement. The third interviewee was affiliated with the Office of the Comptroller General (CGU), the Brazilian federal institution responsible for internal control and integrity oversight within the executive branch. All interviewees possessed substantial professional experience, with a minimum of ten years working in public-sector auditing, investigation, or oversight activities. The interviews were conducted remotely and collectively amounted to approximately eight hours of recorded discussion. To ensure confidentiality and comply with ethical research practices, the identities of the participants are not disclosed.

The interviews followed a semi-structured protocol designed to explore how auditors currently use data and indicators when investigating irregularities in PP. The interview guide included questions addressing five main dimensions: (i) analytical tools currently used to support procurement investigations, (ii) the types of risk indicators or signals commonly associated with irregularities in procurement processes, (iii) challenges related to data access, integration, and quality across government datasets, (iv) institutional and procedural constraints affecting investigative practices, and (v) transparency and traceability requirements for data-driven indicators to be considered credible within auditing and investigative contexts. The semi-structured format allowed interviewees to elaborate on their professional experiences while maintaining comparability across interviews. This approach enabled the identification of both explicit practices and implicit assumptions that shape how auditors interpret risk signals and prioritize investigations.

The analysis of the interviews was conducted using a thematic approach (Mayring, 2021), consolidating the

auditors’ responses into shared analytical dimensions rather than attributing statements to individual participants. The analysis of the interviews resulted in the consolidation of recurrent themes across participants, which are summarized in Table 1. The table presents the analytical dimensions derived from the thematic interpretation of the interviews, alongside representative example questions and their analytical purpose within the study. This thematic consolidation allowed recurring patterns to emerge across the interviews, particularly regarding the practical use of analytical tools, the interpretation of risk indicators, challenges related to data access and integration, institutional constraints affecting investigative processes, and transparency requirements for data-driven indicators. Through this process, the interviews contributed to the identification and refinement of several procurement risk indicators grounded in practitioners’ experience. These indicators complement those derived from the literature and are incorporated into the method instantiation phase presented in Subsection 5.2, where they are operationalized using publicly available government datasets.

Table 1 – Summary of Key Themes Identified in Auditor Interviews

Dimension	Example Question	Purpose
Analytical Tools	What analytical tools are currently used to support procurement investigations?	Understand current analytical capacity
Risk Indicators	What signals or patterns may indicate fraud risks in procurement?	Identify practitioner driven indicators
Data Access	What challenges exist in accessing or integrating procurement datasets?	Identify data integration constraints
Institutional Constraints	What institutional factors affect investigative prioritization?	Understand operational limitations
Transparency	What level of explanation is required for analytical indicators to be trusted?	Understand transparency requirements

Overall, the interviews revealed several gaps in the analytical capabilities and the operational realities of public sector auditing. In particular, auditors highlighted the absence of integrated analytical systems to support fraud detection, the fragmentation and limited interoperability of relevant datasets, and the difficulty of translating analytical signals into legally actionable evidence. These constraints reinforce the need for risk indicators that are transparent, interpretable, and explicitly linked to their underlying data sources and transformation processes. The insights obtained from the interviews, therefore, played a critical role in shaping the design requirements of the proposed method, particularly the emphasis on traceability, data integration, and the explicit modeling of provenance in the construction of procurement risk indicators.

3. Literature Review

To identify and consolidate red flags applicable to PP, we conducted a structured rapid review protocol (Kazi et al., 2021), a type of Systematic Literature Review, to rapid evidence synthesis from the literature. The objective of this review was to systematically examine how red flags, fraud indicators, and risk signals are defined and generated in the PP, and it review was guided by the following research question: *How are red flags and risk indicators defined and applied to identify fraud risks in public procurement?*

We used the following scientific databases to collect the papers: Scopus ¹, ACM ², and IEEE Xplore ³. We used the following search string: (“Red Flags” OR “Fraud indicators” OR “Risk indicators” OR “Risk signals”) AND (“Public Procurement” OR “Electronic Government” OR “e-gov”). Inclusion criteria comprised studies that define, analyze, or generate fraud risk indicators in PP, including relevant systematic reviews and empirical contributions. Exclusion criteria removed studies outside the PP domain, those lacking substantive discussion of risk identification, non-English publications, incomplete records, and duplicates. Fig. 1 summarizes the study selection process, which details the identification, screening, eligibility, and inclusion phases, illustrating the number of records retrieved, excluded, and retained at each stage of the review. A structured mapping between the identified red flags and their supporting evidence from the rapid review is provided in Subsection 5.2, Table 2, enabling traceability of the literature sources underlying each indicator. Additionally, one further study (Velasco et al., 2021) was incorporated through complementary screening procedures due to its high conceptual and empirical relevance to red flag generation in PP.

¹<https://www.scopus.com/home.uri>
²<https://dl.acm.org/>
³<https://ieeexplore.ieee.org/Xplore/home.jsp>

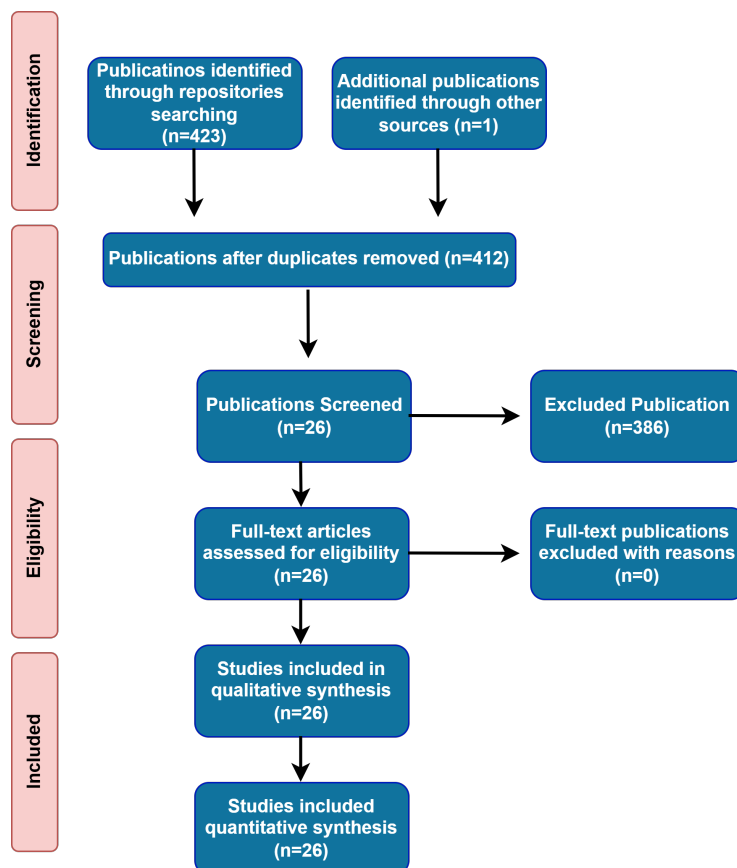


Figure 1 – Information flow through the steps of the review using PRISMA Flow Diagram

The literature review revealed a broad body of research investigating fraud detection and risk signals in PP, highlighting several recurring indicators related to restricted competition, political connections, abnormal bidding patterns, etc. These studies provide valuable conceptual foundations for identifying potential corruption risks and demonstrate the growing use of data driven approaches in procurement oversight. However, the review also exposed important gaps. Many studies focus primarily on statistical detection techniques applied to specific datasets, while offering limited transparency regarding the origin of the data, the construction of the indicators, or the analytical procedures used to derive them. In several cases, datasets are integrated without traceable lineage, indicators are generated without a clear methodological framework, and model outputs are presented without transparent explanations or validation from domain experts. As a result, the analytical processes linking raw government data to risk indicators frequently remain insufficiently documented, making them difficult to reproduce, interpret, or audit in institutional oversight contexts.

4. Research Methodology

This study adopts a DSR approach (Hevner et al., 2004), which is particularly suitable for developing and evaluating artifacts designed to address real-world problems. In this context, the artifact corresponds to a method for constructing open risk indicators for public procurement based on government data and provenance modeling. DSR emphasizes the integration of practical relevance and theoretical rigor through iterative interactions between the problem environment and the knowledge base. The step-by-step methodological workflow adopted in this study is illustrated in Fig. 2.

First, an empirical background was established through semi-structured interviews with public auditors involved in procurement oversight and fraud investigation. This stage corresponds to the relevance cycle in DSR and aims to ground the research problem in real-world practices. The interviews enabled the identification of key challenges, such as data fragmentation, limited interoperability, and the need for transparent and interpretable analytical outputs.

Second, the insights derived from the interviews informed the scope and focus of a structured literature review. This stage corresponds to the rigor cycle, in which existing scientific knowledge is used to understand

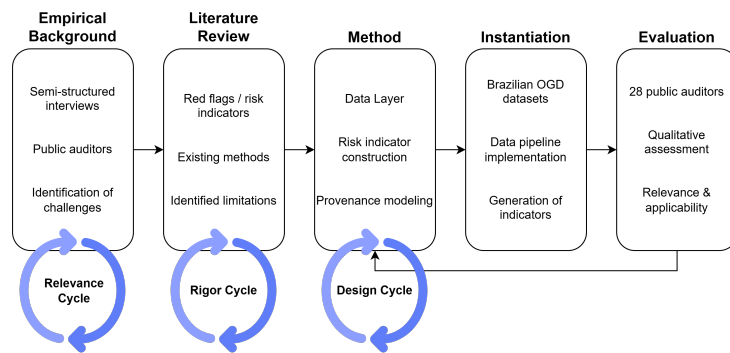


Figure 2 – Research Methodology Based on DSR

how risk indicators (or red flags) have been defined and operationalized in prior studies. The review also helped identify methodological limitations, particularly regarding transparency, traceability, and the lack of explicit data lineage in the construction of indicators.

Third, based on the combination of empirical insights and literature findings, the proposed method was designed. This stage represents the design of the artifact and consists of defining a structured data and analytics pipeline capable of integrating heterogeneous government datasets and transforming them into measurable risk indicators. A key component of the method is the incorporation of a provenance modeling layer, which explicitly represents the relationships between data sources, transformations, and generated indicators, ensuring traceability across the indicator–data–source lifecycle.

Fourth, the method was instantiated in the Brazilian PP context using multiple OGD. This instantiation demonstrates the applicability of the proposed approach in a real-world scenario and enables the generation of a set of risk indicators derived from actual procurement data. Finally, the outputs of the method were evaluated through a qualitative assessment with public auditors. This evaluation stage examines the perceived relevance and practical applicability of both the datasets and the constructed indicators, providing evidence of the method’s usefulness in supporting procurement oversight activities.

5. Method Design and Instantiation

This section presents the proposed method for constructing open risk indicators from Open Government Data (OGD) and its application in a real-world context. Following the research design described in the previous section, the method is first introduced conceptually, detailing its structure, components, and underlying principles. Subsequently, the method is instantiated using Brazilian PP datasets, demonstrating its applicability and enabling the generation of risk indicators based on actual data. This two-step presentation reflects the design and instantiation stages of the study, connecting the methodological proposal with its empirical implementation.

5.1. Method for Constructing Open Risk Indicators from OGD

The aim of our method is to construct open risk indicators for PP using OGD, as well as to structure an integrated PP data environment (PP Data Lakehouse) derived from these datasets. The method was designed to transform heterogeneous government datasets into analytically usable risk indicators while ensuring transparency and traceability in their construction. It combines knowledge elicitation from academic literature and domain experts with a structured data and analytics pipeline that integrates, processes, and operationalizes government datasets into measurable indicators. In addition, the method incorporates a provenance modeling component that documents the data sources, transformations, and analytical steps involved in generating each indicator, enabling explicit traceability of the indicator–data–source relationships.

Fig. 3 illustrates the overall workflow of the proposed method. The input of the method consists of government data sources, which are processed through a structured data and analytics pipeline. The data layer begins with the mapping of available government datasets, followed by data extraction and preparation steps that enable the integration and standardization of heterogeneous sources. The prepared data are then used for the construction of risk indicators. In parallel, a provenance modeling component documents the data

sources, transformations, and analytical steps involved in generating each indicator, ensuring transparency and traceability across the indicator–data–source lifecycle. The outputs of the method include a procurement data environment (PP Data Lakehouse) and a set of traceable risk indicators derived from government data.

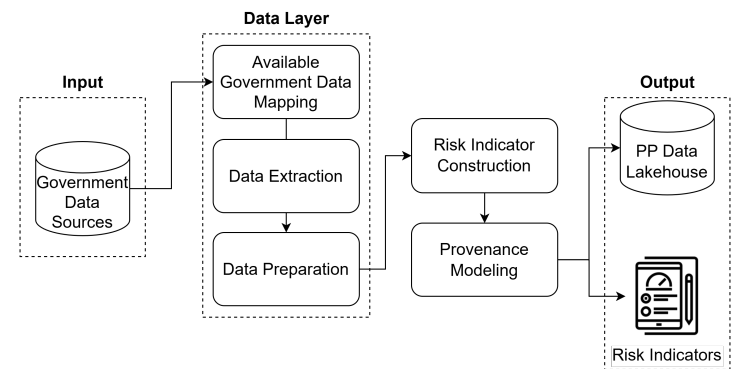


Figure 3 – Proposed Method

5.2. Instantiation our Method

Data sources were selected to ensure reliable and comprehensive coverage of Brazil’s PP processes, prioritizing publicly maintained datasets released under open data policies. The data structure integrates eleven official datasets that cover different stages of the PP process and related administrative domains. The Transparency Portal (D1) ⁴ provides OGD on public expenditures, including detailed payment records from social programs such as Bolsa Familia (D2) ⁵. The Superior Electoral Court (D3) ⁶ dataset contains information on candidates, political parties, campaign donations, and election results. The Government Procurement Portal (CATMAT/CATSER) (D4) ⁷ lists goods and services procured by the federal government, while the ComprasNet Contracts API (D5) ⁸ provides OGD of tenders, contracts, suppliers, and contract values. The National Registry of Legal Entities (CNPJ) (D6) ⁹ offers company registration data and identifiers, and the Accounts System of the Federal Court of Accounts (TCU) (D7) ¹⁰ compiles institutional accounting and audit information. The Registry of Suspended and Debarred Companies (CEIS) (D8) ¹¹ and the National Registry of Punished Companies (CNEP) (D9) ¹² both contain OGD with lists of suppliers sanctioned by oversight bodies. The Transparency Portal – Invoices ¹³ (D10) includes invoice-level OGD on contract execution, and the Safra Guarantee Program (D11) ¹⁴ OGD provides records of financial transfers to rural producers. Collectively, these data sets constitute a comprehensive and auditable basis for analyzing procurement activities using OGD.

Fig. 4 shows that data acquisition was automated through Python scripts within a scalable cloud environment. The architecture is organized into three main layers. In the computing layer, Python and PySpark technologies were used and executed on Amazon EMR to perform extraction, normalization, validation, and identifier harmonization procedures. Raw datasets were first ingested into an Amazon S3 landing zone, where they were preserved in their original format. Subsequently, transformation and integration routines were applied through a structured data engineering pipeline, generating harmonized datasets stored in a refined zone using Parquet format. The final processed datasets constitute the analytical data layer used for red flag generation. All code artifacts were version controlled and managed through a dedicated GitHub repository ¹⁵, forming a distinct code layer that ensures reproducibility, transparency, and traceability across the entire data pipeline.

Table 2 consolidates heuristic indicators from both research and auditing practice, providing a standardized reference for implementation. Within the methodological workflow illustrated in Fig. 3, the indicators presented in Table 2 correspond to the Risk Indicator Construction stage, while the Data Layer introduced in

⁴<https://portaldatransparencia.gov.br/>
⁵<https://portaldatransparencia.gov.br/download-de-dados/bolsa-familia-pagamentos>
⁶<https://dadosabertos.tse.jus.br/dataset/>
⁷<https://www.gov.br/compras/pt-br/aceso-a-informacao/consulta-detalhada/planilha-catmat-catser>
⁸<https://contratos.comprasnet.gov.br/api/docs>
⁹<https://dados.gov.br/dados/conjuntos-dados/cadastro-nacional-da-pessoa-juridica---cnpj>
¹⁰<https://contas.tcu.gov.br/ords/?p=1660:2::NO::>
¹¹<https://portaldatransparencia.gov.br/download-de-dados/ceis>
¹²<https://portaldatransparencia.gov.br/download-de-dados/cnep>
¹³<https://portaldatransparencia.gov.br/download-de-dados/notas-fiscais>
¹⁴<https://portaldatransparencia.gov.br/download-de-dados/garantia-safra>
¹⁵https://github.com/s-fontes/projeto_capivara

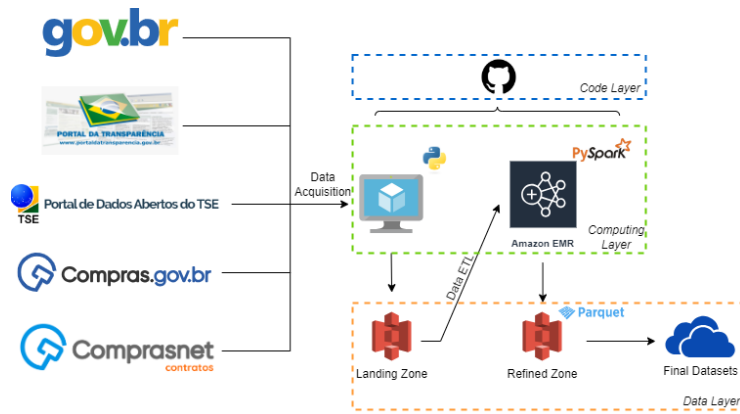


Figure 4 – Data Extraction and Preparation Architecture

the same figure is operationalized through the technical architecture presented in Fig. 4, which details how heterogeneous government datasets are extracted, integrated, and prepared prior to indicator generation. To facilitate interpretation, Table 2 includes references to several Brazilian administrative datasets and registries. Among them, CNPJ (Cadastro Nacional da Pessoa Jurídica) is the Brazilian National Registry of Legal Entities, which uniquely identifies companies; CPF (Cadastro de Pessoas Físicas) is the Brazilian Individual Taxpayer Registry, used to uniquely identify citizens; TSE (Tribunal Superior Eleitoral) is Brazil's Superior Electoral Court, responsible for maintaining official electoral records such as candidates, political parties, campaign donations, and party affiliations; and CNAE (Classificação Nacional de Atividades Econômicas) is the Brazilian National Classification of Economic Activities, which categorizes the primary and secondary economic activities performed by companies. Some red flags reflect specific characteristics of the Brazilian institutional and OGD environment, such as social program registries or national sanction lists, yet their underlying risk logic is grounded in patterns identified in the international literature. Although the empirical operationalization is adapted to the Brazilian data ecosystem, the conceptual references in prior studies often address closely related phenomena (e.g., political connections, competition restriction, sanctioned firms, or conflicts of interest), even when derived from different national contexts. Each indicator therefore corresponds to a specific risk hypothesis that can be empirically tested using open datasets, while remaining theoretically anchored in broader research traditions. The inclusion of the "Calculation" and "Limitations" fields acknowledges that red flags do not constitute evidence of fraud but rather probabilistic risk markers subject to contextual interpretation. This hybrid design, grounded in both empirical and expert knowledge, ensures that the framework captures not only data patterns but also the auditing rationale behind them, reinforcing interpretability and institutional relevance. While identifying red flags establishes measurable risk indicators, ensuring the traceability of their construction is essential to validating analytical results.

Table 2 – Red Flags Mapping to Data Sources, Calculation Procedures, and Identified Limitations

Code	Source	Indicator	Dataset(s)	Description	Calculation	Limitation
F1	(Fazkas & Kocis, 2020), (Velasco et al., 2021)	Bolsa Família Beneficiary Partner	Transparency Portal (Bolsa Família), CNPJ	Partner receives benefits	CPF-CNPJ match, deduplication	Outdated social data, possible false positives.
F2	(Fazkas & Kocis, 2020), (Tishchenko, 2016), Auditor interview	Garantia Safra Beneficiary Partner	Safra Guarantee Program, CNPJ	Partner in the Safra program	CPF-CNPJ match, deduplication	Outdated data, potential inclusion of legitimate beneficiaries.
F3	(Nielsen et al., 2020), (Fazkas et al., 2016), (Fazkas & Kocis, 2020)	Recently Created Company	CNPJ, ComprasNet	New firm before contract	Isnt diff (open-contract)	Legitimate new firms may appear; contract is not considered.
F4	(Licciandrea et al., 2022), (Fazkas & Kocis, 2020)	Low Share Capital vs. Contract Value	CNPJ, ComprasNet	Low capital, high value	Value ratio (capital-contract)	Share capital may be symbolic; low reliability.
F5	(Ferreira et al., 2017), (Fazkas & Kocis, 2020), (Velasco et al., 2021)	Partner with a Political Candidate	TSE, CNPJ	Partner linked to candidate	CPF matches the candidate	Political link does not imply favoritism.
F6	(Fazkas & Kocis, 2020)	Company with Tax Exemptions	Tax Authority, CNPJ	Firm under tax relief	CNPJ match; tax data join	Registry errors may affect classification.
F7	(Martinez-Plumed et al., 2018), (Tishchenko, 2016)	Sanctioned Company	CBS/CNE, ComprasNet	Firm previously sanctioned	ID match; sanction join	Sanctions may be outdated or unrelated.
F8	(Tishchenko, 2016), (Velasco et al., 2021)	Partner as Active Public Servant	Public Servant Registry, CNPJ	Partner is a civil servant	CPF matches the servant list	Some positions lack disclosure; generalization risk.
F9	(Westerski et al., 2021), (Tishchenko, 2016)	Shared Telephone Number	CNPJ	Same phone, multiple firms	Phone ntd; duplicate check	Shared contact data may have legitimate reasons.
F10	(Westerski et al., 2021), (Tishchenko, 2016)	Shared Email Address	CNPJ	Same email, multiple firms	Email ntd; duplicate check	Common emails do not necessarily indicate irregularity.
F11	(Westerski et al., 2021), (Brandão et al., 2024), (Fuentes et al., 2025), (Martinez-Plumed et al., 2018), (Modrutan et al., 2021)	Shared Business Address	CNPJ	Same address multiple firms	Address ntd; geocode; match	Shared locations may be coincidental or legitimate.
F12	(Nielsen et al., 2020), (Licciandrea et al., 2022), (Fazkas et al., 2016), (Ferreira et al., 2017), (Ulloa et al., 2025), (Fazkas & Wachs, 2020), (Decarolis & Giorgiannini, 2022), (Tish & Hajdu, 2025), (Fazkas et al., 2022), (Gualdi & Del Sarto, 2024), (Fazkas et al., 2024), (Tatrai & Nemeth, 2018), (Goshank, 2020), (Fuentes et al., 2025), (Modrutan et al., 2021)	Partners in Multiple Companies	CNPJ	Partner owns several firms	Partner-firm network link	Legal corporate structures may justify overlap.
F13	(Nielsen et al., 2020), (Licciandrea et al., 2022), (Fazkas et al., 2016), (Ferreira et al., 2017), (Ulloa et al., 2025), (Fazkas & Wachs, 2020), (Decarolis & Giorgiannini, 2022), (Tish & Hajdu, 2025), (Fazkas et al., 2022), (Gualdi & Del Sarto, 2024), (Fazkas et al., 2024), (Tatrai & Nemeth, 2018), (Goshank, 2020), (Fuentes et al., 2025), (Modrutan et al., 2021)	Contracts Obtained Through Bidding	ComprasNet	Contracts via bidding	Classify; ratio (bidding)	Competitive processes may still present risks.
F14	(Nielsen et al., 2020), (Licciandrea et al., 2022), (Fazkas et al., 2016), (Ferreira et al., 2017), (Ulloa et al., 2025), (Fazkas & Wachs, 2020), (Decarolis & Giorgiannini, 2022), (Tish & Hajdu, 2025), (Fazkas et al., 2022), (Gualdi & Del Sarto, 2024), (Fazkas et al., 2024), (Tatrai & Nemeth, 2018), (Goshank, 2020), (Fuentes et al., 2025), (Modrutan et al., 2021)	Contracts Obtained Without Bidding	ComprasNet	Direct award contracts	Classify; ratio (no bidding)	Legal exemptions are possible; it depends on the contract type.
F15	(Fazkas et al., 2022), (Fazkas et al., 2024), (Tishchenko, 2016)	High Contract Value per Capita	ComprasNet, IBGE	High value per citizen	Normalize; join pop; per capita	Regional or budget differences may distort ratios.
F16	(Nielsen et al., 2020), (Licciandrea et al., 2022), (Fazkas et al., 2016), (Ulloa et al., 2025), (Fazkas et al., 2022), (Fazkas et al., 2024)	High Contract Value Relative to Agency Budget	ComprasNet, Agency Budgets	Value exceeds budget	Normalize; join budget; ratio	The agency budget scale may bias interpretation.
F17	(Velasco et al., 2021)	Inactive CNPJ	CNPJ	Company registration inactive	Check CNPJ status	Registry inaccuracies may lead to misclassification.
F18	(Tishchenko, 2016)	Tax Regime Incompatible with Contract Value	Tax Authority, ComprasNet	Isn type mismatch	Match tax regime; compare	Registration or reporting errors are possible.
F19	(Velasco et al., 2021)	Company Donations to Political Campaigns	TSE (Donations), CNPJ	Firm donates to campaign	CNPJ match; donation join	Legal donations; no direct fraud evidence.
F20	(Modrutan et al., 2021), (Velasco et al., 2021)	Partner Donations to Political Campaigns	TSE (Donations), CNPJ	Partner donates to the campaign	CPF match; donation join	Small donations are often irrelevant; low risk.
F21	(Modrutan et al., 2021), (Velasco et al., 2021)	Partner Affiliations with Political Party	TSE (Party Membership), CNPJ	Partner party member	CPF match; party join	Common membership is not necessarily irregular.
F22	(Modrutan et al., 2021), (Velasco et al., 2021)	Company Services to Electoral Campaigns	TSE (Campaign Services), ComprasNet	Firm served campaigns	CNPJ match; campaign join	Legitimate service provision may occur.
F23	Auditor interview	Ineligible Company (CGU)	CGU Ineligibility List	CGU ineligible supplier	CNPJ match; CGU join	The record may be outdated or under appeal.
F24	Auditor interview	Ineligible Company (TCU)	TCU Ineligibility List	TCU ineligible supplier	CNPJ match; TCU join	Possible duplication or outdated sanction.
F25	Auditor interview	Punished Company (CGU)	CGU Sanctions Database	CGU-sanctioned firm	CNPJ match; CGU punishment	Old or unrelated penalties may persist.
F26	(Ulloa et al., 2025), (Fuentes et al., 2025), (Martinez-Plumed et al., 2018)	CNAE Mismatch with Contract	CNPJ, ComprasNet	Activity not matching contract	CNAE-contract text match	CNAE codes may be generic or misclassified.

We used Table 2 as input to the provenance modeling of the constructed risk indicators. Table 2 specifies what each risk indicator represents (risk hypothesis), which datasets are required, and how the indicators are calculated. The code and datasets used to generate these indicators are available at Github. To address the need for traceability, the method incorporates a provenance model that formally represents the expected workflow of indicator generation and links each computed risk indicator to its underlying data sources, transformation steps, and responsible actors or systems. Fig. 5 presents the provenance graph documenting the derivation of

RedFlag_001 by explicitly connecting entities (e.g., procurement and supplier records), activities (e.g., registration, joins, filtering, and indicator computation), and agents (e.g., scripts, triggers, or systems executing the pipeline). In this way, provenance complements the indicator definitions in Table 2 by providing an auditable record of the indicator–data–source lifecycle, enabling stakeholders to reconstruct, inspect, and justify the analytical pathway that produced each risk indicator.

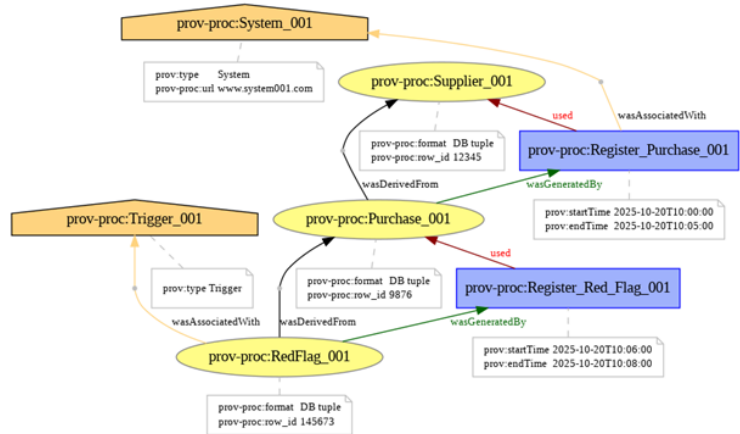


Figure 5 – Data provenance model for fraud risk indicators

6. Evaluation

This section presents the qualitative evaluation of the proposed datasets and red flag indicators based on a structured questionnaire administered to 28 public auditors. The instrument was designed to capture practitioners’ perspectives on the practical validity of each indicator and the adequacy of the datasets supporting their construction. Beyond reporting levels of agreement, the objective is to examine how these elements are perceived within real oversight contexts, considering relevance, completeness, reliability, and operational feasibility. The analysis combines quantitative assessments with qualitative feedback, allowing for a nuanced interpretation of consensus and divergence across responses.

The evaluation focused on two main analytical criteria: (i) the perceived relevance of the datasets for fraud detection activities and (ii) the practical applicability of the proposed risk indicators in procurement oversight contexts. These criteria were assessed through both closed-ended and open-ended questions. The quantitative responses were consolidated into the visual summaries presented in Fig. 6 and Fig. 8, while the qualitative comments were used to contextualize and interpret the observed response patterns.

Participants were asked to assess the relevance of various public datasets in the context of fraud detection. They were presented with a list of datasets and, for each one, indicated whether they considered it relevant for detecting fraud, with an optional follow-up question allowing them to justify their responses. A final open-ended question invited participants to suggest additional public datasets they deemed relevant to the study. This structure provided insights into the practical applicability of existing datasets as well as potential gaps in the resources available for fraud detection. Each dataset-related question included three response options: “Yes,” “No,” and “I do not know or prefer not to answer,” allowing participants to express agreement, disagreement, or uncertainty, thereby ensuring comprehensive and nuanced feedback. Fig. 6 summarizes the responses to each dataset-related question, where “Q” denotes the questionnaire item and “D1,” “D2,” and subsequent labels refer to the respective datasets evaluated.

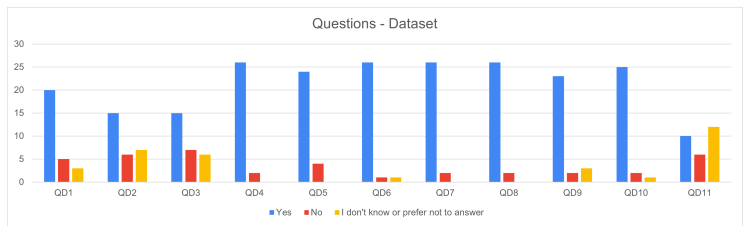


Figure 6 – Answers to datasets questions

Participants were also asked to assess the practical relevance of each proposed red flag for detecting fraud in PP. This step aimed to capture insights into the applicability of the indicators in day to day auditing activities

and to identify potential gaps or adjustments needed to refine them. By adopting a practice oriented perspective, this assessment also explored auditors' perceptions regarding the extent to which the red flags reflect patterns and behaviors typically associated with fraudulent practices. Fig. 7 and Fig. 8 provides a consolidated overview of the responses to each red flag evaluated, where "Q" denotes the questionnaire item and "F1," "F2," and subsequent labels refer to the respective red flags assessed.

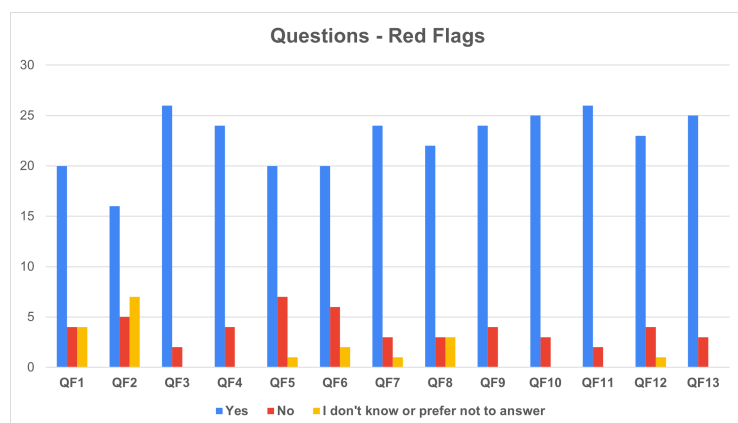


Figure 7 – Red Flags Questions 1-13

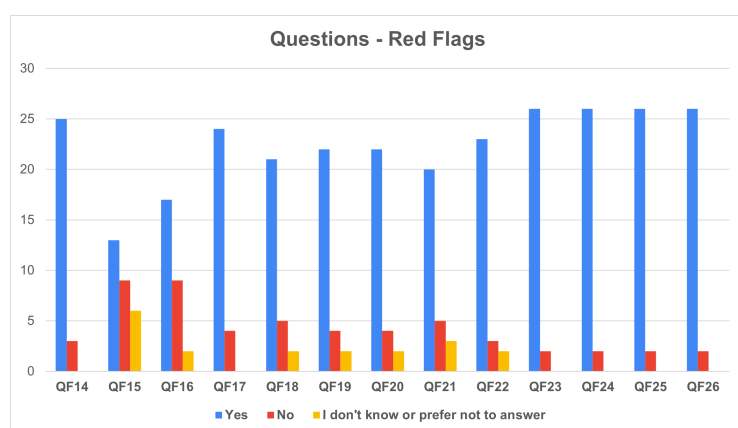


Figure 8 – Red Flags Questions 14-26

Due to space constraints, we present a summary of the main recurrent observations raised by the participants in open-ended comments in what follows. Overall, the auditors' comments indicate a strong consensus regarding the relevance of the proposed datasets for detecting potential signs of fraud, particularly when used in an integrated manner. Respondents emphasized the analytical value of cross-referencing databases such as Bolsa Família, the Superior Electoral Court (TSE), and the National Registry of Legal Entities (CNPJ) to identify possible straw men, political connections, income inconsistencies, and registration anomalies. Likewise, procurement-related datasets, including contract records, suppliers, and invoices, were considered useful for assessing operational capacity, contractual history, and pricing patterns. Databases of ineligible or sanctioned companies were also highlighted as essential mechanisms to prevent irregular contracting. At the same time, auditors pointed out several practical limitations, including outdated information, datasets not specifically designed for procurement oversight, technical barriers to data access, and the need for greater data standardization and training. Overall, respondents emphasized that the greatest analytical value lies in the systematic and structured integration of these heterogeneous information sources.

7. Discussions

The findings indicate strong practitioner alignment with the datasets and risk indicators proposed in this study, as most of them were considered relevant and applicable by the participating auditors. At the same time, the results reinforce that such indicators should be used strictly as decision support instruments rather than as automated evidence of wrongdoing, assisting auditors in prioritizing cases while preserving human judgment and institutional procedures in investigative processes. This perspective is consistent with prior research emphasizing the role of data-driven approaches as supportive tools rather than substitutes for expert judgment in public sector auditing (Felizzola et al., 2024; Soylyu et al., 2022).

Beyond their operational use, the study positions data provenance as a fundamental mechanism for ensuring traceability and accountability in the construction of open risk indicators within OGD ecosystems. By explicitly documenting the relationships between data sources, transformations, and generated indicators, provenance enables the reconstruction of the analytical pathway that produces each indicator, strengthening transparency and institutional legitimacy. This finding directly addresses limitations identified in prior studies, which often rely on statistical or machine learning approaches without explicitly documenting data lineage or analytical processes (Lisciandra et al., 2022; Modrušan et al., 2021). In this sense, the contribution of this study lies not only in identifying relevant risk indicators, but in formalizing their construction as traceable and auditable analytical artifacts, thereby extending existing research on OGD in PP beyond data availability and analytical applications.

Finally, the results also highlight the importance of data integration as a central enabler of analytical capability, reinforcing findings from previous studies that emphasize the need for combining heterogeneous datasets to detect complex patterns of irregularities. However, unlike prior work that often assumes the availability of integrated data environments, this study explicitly incorporates data integration as part of the methodological design, thereby addressing a critical gap between theoretical approaches and practical implementation in public sector contexts.

8. Conclusion and Future Research

This study proposes a method for constructing open risk indicators for PP using OGD. A provenance modeling component, inspired by the W3C PROV-DM standard, documents the relationships between data sources, transformations, and generated indicators to guarantee transparency and traceability. The method was instantiated in the Brazilian context, resulting in the construction of 26 risk indicators derived from literature and auditing practice, and was validated through a survey with 28 public procurement auditors. By doing so, the study addresses an important gap in the data-driven PP analysis, as in the literature, datasets are often integrated without traceable lineage, indicators are constructed without clear methodological frameworks, beyond the model outputs being produced without transparent explanations or expert validation. The proposed method contributes by providing a structured and reproducible approach for constructing risk indicators directly from government data while using provenance modeling as a governance mechanism to document the indicator–data–source lifecycle.

As our qualitative evaluation involved a limited number of auditors and was conducted exclusively within the Brazilian PP context, which may constrain generalizability to other jurisdictions. In this way, we have been working on an attemptive to analyze the applicability of our methods in other scenarios, such as Colombia and Europe, via partnerships. Further studies may also explore the broader use of provenance modeling for documenting analytical processes and decision support systems in the digital public service delivery when traceability and other data qualities are required.

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Declarations

Conflict Of Interest (COI)

There is no conflict of interest.

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